

AI-Augmented MBSE for Ontology-Driven Digital Twin Systems Interoperability

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Abstract. Digital Twins (DTs) are essential in modern Cyber-Physical Systems (CPS) for real-time monitoring, representation, and optimization of complex assets. However, semantic interoperability across heterogeneous subsystems remains challenging due to the complexity of formal modeling languages like SysML. This paper introduces a novel AI-assisted methodology bridging natural language system specifications and formal, machine interpretable models for DTs. Our contributions include: (1) a generative AI framework based on LLaMA3 for automatic SysML diagram generation, (2) a rule based pipeline transforming diagrams into semantically enriched ontologies, and (3) an ontology alignment mechanism enabling interoperability across DT systems. Validation on MAGPIE project use cases: Vessel Traffic Monitoring and Port Equipment Maintenance, demonstrates the approach’s effectiveness in model generation, ontology integration, and reasoning, highlighting the potential of combining MBSE with large language models to advance semantic interoperability in multi domain digital ecosystems.

Keywords: Digital Twin, Model-Based Systems Engineering (MBSE), Large Language Models (LLMs), Ontology Alignment, SysML.

1 Introduction

Digital Twins (DTs) are increasingly adopted to enable data-driven analysis and lifecycle management of complex cyber-physical systems. Yet, ensuring semantic interoperability across heterogeneous DTs remains a persistent challenge, as most models and data sources are defined using distinct terminologies and engineering tools. Model-Based Systems Engineering (MBSE) offers a structured foundation for model creation, but building and maintaining formal models requires expert knowledge.

To address this gap, this paper proposes an AI-augmented MBSE framework that leverages Large Language Models (LLMs) to automatically generate SysML artifacts from natural language specifications, transform them into OWL-based ontologies, and perform ontology alignment to support cross-system reasoning. The approach is implemented within the European MAGPIE project [1] and evaluated on realistic port logistics scenarios.

The main contributions are: (1) a GenAI-assisted pipeline connecting textual requirements, SysML diagrams and ontological representations; (2) a rule based SysML-to-OWL transformation enabling semantic interoperability; and (3) an experimental validation demonstrating feasibility on industrial DT systems.

Our evaluation demonstrates that the proposed framework greatly improves system modeling accessibility, allowing users without MBSE expertise to generate models with semantic accuracy close to that of experts. By automating model creation and semantic integration, it enables non specialists to achieve interoperability across complex systems, a process typically requiring advanced formal modeling skills.

Despite these contributions, a rigorous validation is required to assess the effectiveness of AI-assisted MBSE for Digital Twin interoperability. Therefore, this paper addresses the following research questions:

RQ1: To what extent can Large Language Models generate syntactically and semantically valid SysML models from natural language system descriptions?

RQ2: How accurately can SysML models be transformed into ontologies while preserving structural and behavioral semantics?

RQ3: Does the proposed ontology alignment approach effectively improve semantic interoperability across heterogeneous Digital Twin systems?

To answer these questions, we conduct a systematic experimental evaluation on two industrial inspired Digital Twin subsystems from the MAGPIE project, comparing automatically generated models against expert defined references and assessing model validity, transformation correctness, and alignment quality using quantitative and qualitative metrics.

The remainder of the paper is organized as follows. Section 2 briefly describes MBSE and SysML, Ontologies and Interoperability in DT and Generative AI and Large Language Models. Section 3 discusses related work. Section 4 describes the proposed approach and details its steps. Section 5 presents an illustration of the proposed methodology. Finally, Section 6 reports our conclusions and presents future work.

2 Background

Model-Based Systems Engineering (MBSE) structures system specification through formal modeling languages such as SysML [9, 2], ensuring consistency between requirements, structure and behavior. However, maintaining semantic interoperability across heterogeneous MBSE tools and Digital Twin (DT) [3] platforms remains complex. Ontologies (RDF/OWL) offer a shared semantic layer for representing domain concepts and relations, while industrial frameworks such as RAMI 4.0 [15] and the Asset Administration Shell (AAS)[14] provide standardized asset descriptions and lifecycle views. These initiatives, although essential, do not automate the translation of informal textual knowledge into interoperable models.

Recent progress in Large Language Models (LLMs) [5, 7] opens new possibilities for bridging this gap. LLMs can interpret unstructured natural language

specifications and generate structured engineering artifacts. Building on this capability, our work combines generative AI and MBSE to enable end-to-end semantic alignment of DT systems, from text input to ontology based reasoning.

3 Related works

Previous research has explored several directions to improve semantic interoperability across Digital Twin (DT) systems. Traditional MBSE frameworks focus on the consistency and traceability of engineering models, but they still rely on manual specification and provide limited support for cross domain semantic reasoning. Ontology based approaches (e.g., OWL, RDF) [11, 4] have been proposed to formalize system knowledge and enable reasoning, yet most require expert driven model construction and alignment.

More recently, the combination of Large Language Models (LLMs) and knowledge graphs has opened new opportunities for automated model generation and semantic enrichment. Luo et al. [16] introduced Graph Constrained Reasoning, enforcing logical consistency during LLM reasoning on RDF graphs. Sequeda et al. [17] demonstrated that knowledge graphs can improve LLM accuracy in structured data querying, while Xia et al. [14] applied LLMs to automatically generate components of the Asset Administration Shell (AAS) for Industry 4.0 interoperability.

Compared with these efforts, our approach addresses an upstream stage of system engineering by integrating LLM-based model generation directly within the MBSE workflow. Instead of reasoning on existing graphs, it automatically creates SysML artifacts from textual specifications, transforms them into OWL ontologies, and performs ontology alignment to ensure end-to-end semantic interoperability between heterogeneous DT systems.

Discussion and Positioning: Although the above approaches demonstrate the potential of combining MBSE, ontologies, and Large Language Models, most of them focus either on reasoning over existing knowledge graphs or on qualitative demonstrations of feasibility. In particular, limited attention is given to the systematic evaluation of the end-to-end process, from natural language specifications to interoperable ontological representations.

Existing works rarely assess the syntactic and semantic validity of automatically generated SysML models, nor do they quantitatively evaluate ontology transformation correctness or alignment quality across heterogeneous Digital Twin systems. As a result, the practical reliability and reproducibility of these approaches remain insufficiently validated.

In contrast, our work explicitly targets this gap by providing an integrated LLM assisted MBSE workflow accompanied by a structured experimental evaluation. The proposed framework is assessed using well defined metrics for model validity, ontology consistency, and alignment effectiveness, enabling a transparent comparison with expert defined references and ensuring its applicability in real world Digital Twin environments.

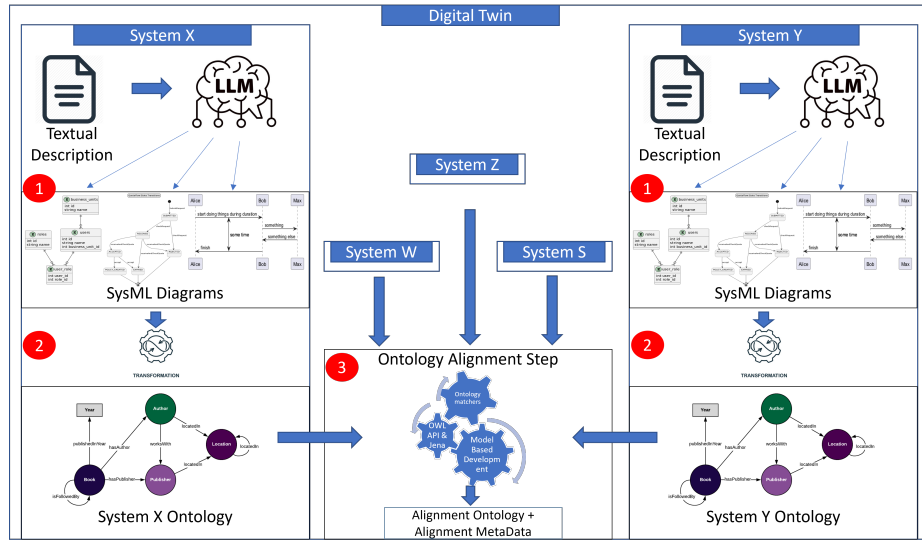


Fig. 1. Proposed methodology architecture

4 Proposed Methodology

The proposed architecture, illustrated in Figure 1, bridges the gap between informal textual descriptions and formal model-based interoperability in Digital Twin (DT) environments. While ontologies enable semantic interoperability, their manual creation and alignment limit scalability and agility. To address this, our methodology automates both model generation and ontology construction, creating a dynamic and adaptable semantic layer grounded in MBSE principles.

The workflow allows domain experts to describe system functionalities in natural language, which are then automatically converted into SysML diagrams by a Large Language Model (LLM). The resulting SysML artifacts are transformed into OWL ontologies that can be aligned across systems to enable reasoning and semantic consistency.

Overall, the architecture implements a complete end-to-end pipeline—from textual input to ontology alignment—comprising three main stages, as highlighted in Figure 1.

1. Text-to-SysML Generation using LLaMA3
2. SysML-to-Ontology Transformation
3. Ontology Alignment for Interoperability

Finally, reasoning and validation services are applied to the generated ontologies and their alignments to perform tasks such as consistency checking, instance verification, and ensuring interoperability.

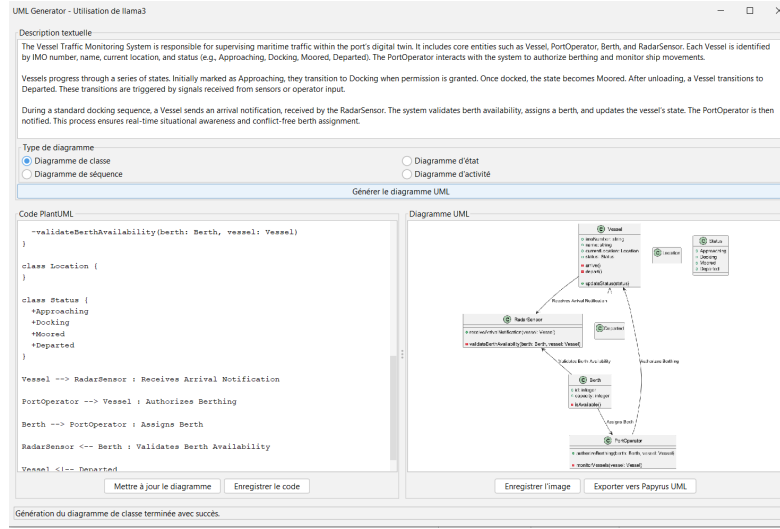


Fig. 2. Text-to-SysML generation tool supporting four SysML diagram.

4.1 Text-to-SysML Generation using LLaMA3

This step aims to automatically generate SysML models from natural language system specifications using a Large Language Model (LLM). The objective is to translate informal textual descriptions into structured SysML artifacts while preserving both structural and behavioral semantics.

The proposed approach relies on the LLaMA3 model to generate SysML diagrams expressed in PlantUML syntax as presented in Figure 2. The generated PlantUML code is automatically validated and imported into Eclipse Papyrus, ensuring conformance with the SysML 1.6 metamodel and enabling further model manipulation. The current implementation supports four SysML diagram types: Block Definition Diagrams (BDD), State Machine Diagrams, Activity Diagrams, and Sequence Diagrams.

To guide the LLM toward syntactically valid and semantically meaningful outputs, domain specific prompt templates are defined for each supported diagram type. These templates combine contextual priming, explicit modeling constraints, and PlantUML syntax guidance. This lightweight prompt engineering strategy avoids model fine tuning while ensuring consistent diagram generation across different system descriptions.

The quality of the generated SysML diagrams is systematically evaluated in Section 5.2 using quantitative and qualitative metrics.

4.2 SysML-to-Ontology Transformation

This stage implements the Model-to-Model (M2M) transformation that converts SysML artifacts into ontological representations following the principles of

Model-Driven Engineering (MDE). The process is implemented in Java using the OWL API [11], where the source metamodel is the SysML specification [9] and the target is the Ontology Definition Metamodel (ODM) [10].

While existing APIs provide generic ontology manipulation capabilities, they do not directly address the semantic preservation required for translating MBSE artifacts. To bridge this gap, our approach defines a domain specific set of SysML-to-OWL transformation rules (see Table 1) that explicitly maintain both structural and behavioral semantics of Digital Twin systems. The mapping covers key SysML constructs—blocks, ports, associations, states, transitions, and messages—ensuring that dynamic aspects such as state changes and message exchanges are faithfully represented in OWL.

Each SysML diagram type is handled through a dedicated subset of rules. For example, associations between blocks are mapped to OWLObjectProperty instances with Domain and Range corresponding to the OWL classes derived from the connected blocks. Similarly, state-machine transitions are converted into object properties linking source and target states, while sequence diagram messages are represented as OWL classes with corresponding send/receive events and “CoveredBy” relationships.

This formalization establishes a reproducible bridge between model-based design and semantic reasoning, enabling automated ontology generation from SysML models and supporting interoperable Digital Twin engineering.

To ensure the correctness of the generated ontologies, the transformation process is validated using OWL reasoning services. Logical consistency, class satisfiability, and property coherence are checked using standard OWL reasoners, ensuring that the structural and behavioral semantics encoded from SysML models do not introduce contradictions or unsatisfiable concepts.

4.3 Ontology Alignment for Interoperability

Once ontologies are generated from heterogeneous Digital Twin (DT) systems, the proposed approach aims to enable semantic interoperability by aligning and merging these ontologies into a unified semantic model. Building upon our previous work [4], the approach follows a Model-Driven Engineering (MDE) perspective and is structured into two main phases, as illustrated in Figure 3.

Ontology Alignment. In the first phase, semantic correspondences between entities from multiple ontologies are automatically identified using ontology matching techniques. The Alignment API [12] is employed with complementary matching strategies combining lexical, syntactic, structural, and semantic similarities, including exact name matching, edit distance based matching, substring matching, structural comparison, and WordNet based semantic similarity. The resulting alignments establish explicit correspondences between semantically related concepts across DT systems.

Merged Ontology Refactoring. In the second phase, aligned ontologies are merged and refactored to produce a coherent and unified ontology. Original IRIs are re-

Table 1. Translation rules from SysML to Ontology concepts

Rule	Translation of	SysML concepts		Ontology concepts	
		Element	Constraint	Element	Constraint
1	Package Diagram	PD	PD is a Package Diagram	PD_Class	PD_Class $\in$ OWLClass
2	Block	B	$B \in \text{SysML_Block}$	B_Class	B_Class $\in$ OWLClass $\wedge$ B_Class subClassOf PD_Class
3	Blocks Associations	B_Asso	$B_Asso \in \text{SysML_Associations}$	B_AOP	B_AOP $\in$ OWLObjectProperty $\wedge$ Domains (B_AOP)= source(B_Asso) $\wedge$ Ranges (B_AOP)= target(B_Asso)
4	State-machine of a block	SM, B	$SM \in \text{SysML_State-machine}$ SM is associated to B	SM_Class	SM_Class $\in$ OWLClass $\wedge$ SM_Class subClassOf B_Class
5	State-machine States	$S_1, S_2, \dots S_n$ SM	$S_i \in \text{SysML_States}$ $SM \in \text{SysML_State-machine}$ S_i is a state of SM	$S_Class_1, S_Class_2, \dots S_Class_n$	$S_Class_i \in$ OWLClass $\wedge$ S_Class_i subClassOf SM_Class
6	State-machine Transitions	$T_1, T_2, \dots T_n$ SM	$T_i \in \text{SysML_Transitions}$ $SM \in \text{SysML_State-machine}$ T_i is a transition of SM, T_i is a transition between 2 states S_x and S_y	$T_Class_1, T_Class_2, \dots T_Class_n$	$T_Class_i \in$ OWLObjectProperty $\wedge$ Domains (T_Class _i)= $S_x \wedge$ Ranges (T_Class _i)= S_y
7	Activity	ACT	$ACT \in \text{SysML_Activity}$	ACT_Class	ACT_Class $\in$ OWLClass $\wedge$ SM_Class subClassOf PD_Class
8	Activity actions	$A_1, A_2, \dots A_n$ ACT	$A_i \in \text{SysML_Action}$ $ACT \in \text{SysML_Activity}$ A_i is an action of ACT	$A_Class_1, A_Class_2, \dots A_Class_n$	$A_Class_i \in$ OWLClass $\wedge$ A_Class_i subClassOf ACT_Class
9	Activity Transitions	$TA_1, TA_2, \dots TA_n$ ACT	$TA_i \in \text{SysML_Transitions}$ $ACT \in \text{SysML_Activity}$ T_i is a transition of ACT, TA_i is a transition between 2 actions A_x and A_y	$TA_Class_1, TA_Class_2, \dots TA_Class_n$	$TA_Class_i \in$ OWLObjectProperty $\wedge$ Domains (TA_Class _i)= $A_x \wedge$ Ranges (TA_Class _i)= A_y
10	Sequence Diagram Interaction	INT	$INT \in \text{SysML_Interaction}$	INT_Class	INT_Class $\in$ OWLClass $\wedge$ INT_Class subClassOf PD_Class
11	Lifeline of a Sequence Diagram associated to a block	L, B	$L \in \text{SysML_Lifeline}$ L is an instance of block B, B IS already been translated	L_Ind	L_Ind $\in$ OWLIndividual $\wedge$ Types (L_Ind)=B_Class $\wedge$ ObjectPropertyAssertions (L_Ind)= [Messages CoveredBy L]
12	Sequence Diagram Message	M	$M \in \text{SysML_Message}$	M_Class	M_Class $\in$ OWLClass $\wedge$ M_Class subClassOf INT_Class

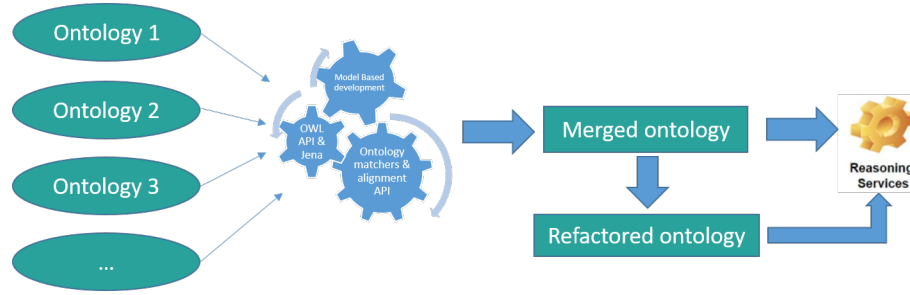


Fig. 3. Multimodal core ontology generation through ontology alignment and refactoring

placed with domain specific IRIs, and bridging axioms are automatically generated using the AROM approach [13]. This refactoring process ensures semantic continuity, removes redundancy, and results in a standalone ontology with a consistent namespace suitable for cross domain reasoning.

The proposed framework extends standard ontology alignment mechanisms by integrating an automated refactoring layer that transforms alignment results into a semantically consistent merged ontology. Reasoning services are then applied to support consistency checking and instance validation across aligned Digital Twin systems.

The effectiveness of the ontology alignment and refactoring process is quantitatively evaluated in Section 5 using standard precision, recall, and F1-score metrics.

5 Experimental Evaluation

This section presents a systematic experimental evaluation of the proposed AI-augmented MBSE framework. The objective is to assess the effectiveness of each stage of the pipeline, from natural language specifications to interoperable ontological representations, in accordance with the research questions introduced in Section 1.

5.1 Experimental Setup

The experimental evaluation is conducted on two industrial inspired Digital Twin subsystems developed within the MAGPIE project: the Vessel Traffic Monitoring System (VTMS) and the Port Equipment Digital Maintenance (PEDM) system. These systems were selected due to their heterogeneous structures, distinct operational behaviors, and interoperability requirements.

For each subsystem, a set of natural language system descriptions was provided as input to the proposed pipeline.

- **Vessel Traffic Monitoring System (VTMS):** VTMS supervises maritime traffic, managing entities such as Vessel, PortOperator, Berth, and RadarSensor. Vessels progress through states (Approaching, Docking, Moored, Departed) based on sensor signals and operator input. The system ensures real time situational awareness and conflict free berth assignment by validating berth availability, updating vessel states, and notifying the PortOperator.
- **Port Equipment Digital Maintenance System (PEDM):** PEDM manages the lifecycle of port equipment, including Cranes, AGVs, and Sensors. Equipment states (Operational, UnderMaintenance, Faulty) change according to sensor alerts and maintenance interventions. The system generates alerts, updates equipment status, and supports technicians in restoring operational conditions, ensuring continuous equipment monitoring and maintenance.

The LLaMA3 large language model was used without fine tuning via the Ollama framework to automatically generate SysML artifacts.

The current implementation of the tool supports four SysML diagram types only: Block Definition Diagrams (BDD), State Machine Diagrams, Activity Diagrams, and Sequence Diagrams. These four diagram types constitute the complete scope of the experimental evaluation.

Expert defined SysML models and ontologies were independently created and used as reference baselines. The evaluation focuses on three aspects: (i) the quality of the generated SysML diagrams, (ii) the correctness and consistency of the SysML-to-ontology transformation, and (iii) the effectiveness of ontology alignment in supporting semantic interoperability.

5.2 Evaluation of Text-to-SysML Generation

This experiment evaluates the ability of the proposed framework to generate valid and accurate SysML models from natural language specifications, thereby addressing RQ1. The evaluation focuses exclusively on the four SysML diagram types generated by the tool: BDD, State Machine, Activity, and Sequence Diagrams.

Three complementary evaluation criteria are applied, as defined in Section 4.1.1. Syntactic validity is measured by verifying whether the generated PlantUML code can be successfully parsed and imported into Eclipse Papyrus without errors. Structural completeness is computed by comparing the generated diagrams against expert defined reference models. For each diagram type, completeness is measured independently as the ratio of correctly identified modeling elements (e.g., blocks, states, actions, messages, and relationships) to the total number of relevant elements in the reference model. The final completeness score is obtained by averaging the results across the four diagram types.

Semantic correctness is assessed by two domain experts, who independently evaluate whether the generated diagrams accurately capture the intended system behavior described in the textual specifications. A five point Likert scale is used, and the reported score corresponds to the average expert rating.

Table 2. Evaluation results of Text-to-SysML generation

Metric	VTMS PEDM	
Syntactic validity (%)	92	89
Structural completeness	0.87	0.83
Semantic correctness (/5)	4.2	4.0

The results presented in Table 2 show that the proposed approach produces syntactically valid SysML diagrams in the majority of cases. Structural completeness remains consistently high across both systems, indicating that the LLM is able to correctly identify most relevant system entities and relationships. Observed semantic deviations were primarily related to implicit domain assumptions that were not explicitly stated in the natural language inputs.

5.3 Evaluation of SysML-to-Ontology Transformation

This experiment addresses RQ2 by evaluating the correctness and consistency of the SysML-to-ontology transformation. The OWL ontologies generated from the four SysML diagram types were analyzed using OWL reasoning services to verify logical consistency, class satisfiability, and property coherence.

For both VTMS and PEDM, the generated ontologies were successfully validated using standard reasoners, confirming the absence of unsatisfiable classes or contradictory axioms. In addition, the size and structural richness of the ontologies were analyzed to ensure that both structural and behavioral semantics encoded in the SysML models were preserved.

Table 3. Ontology transformation results

System	#Classes	#Properties	Consistent
VTMS	134	52	Yes
PEDM	118	47	Yes

These results presented in Table 3 confirm that the proposed transformation rules provide a reliable mapping from SysML artifacts to ontological representations, enabling formal reasoning while preserving the original system semantics.

5.4 Evaluation of Ontology Alignment

The third experiment evaluates the effectiveness of the ontology alignment process in improving semantic interoperability across heterogeneous Digital Twin systems, addressing RQ3. Automatically generated alignments were compared against a manually curated reference alignment created by domain experts.

Alignment quality is measured using standard precision, recall, and F1-score metrics. The proposed alignment mechanism is compared with baseline matching strategies based on exact name matching and edit distance similarity.

Table 4. Ontology alignment performance

Method	Precision	Recall	F1-score
NameEqAlignment	0.62	0.45	0.52
EditDistAlignment	0.70	0.61	0.65
Proposed approach	0.81	0.74	0.77

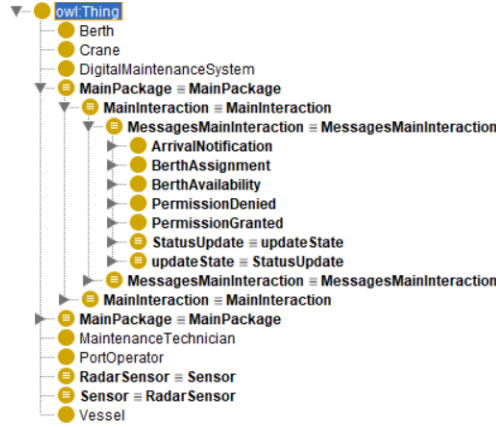


Fig. 4. Merged ontology

Figure 4 illustrates an excerpt of the merged ontology obtained after alignment, highlighting the unification of semantically related concepts across heterogeneous Digital Twin systems.

The results presented in Table 4 demonstrate that the proposed ontology alignment approach significantly outperforms baseline strategies, enabling more accurate semantic correspondences and supporting interoperable reasoning across Digital Twin systems.

5.5 Discussion and Threats to Validity

The experimental results demonstrate the feasibility and effectiveness of the proposed AI-augmented MBSE framework in automating the generation of SysML diagrams, transforming them into ontologies, and aligning heterogeneous Digital Twin systems. The evaluation confirms that Large Language Models can

substantially reduce modeling effort while maintaining a high level of semantic accuracy.

Nevertheless, several threats to validity must be acknowledged. First, semantic correctness evaluation relies partially on expert judgment, which may introduce subjective bias. Second, the evaluation is limited to four SysML diagram types supported by the current implementation. Finally, the experiments were conducted on two industrial inspired systems within the port logistics domain, which may limit generalizability. Future work will address these limitations by extending the evaluation to additional domains, diagram types, and larger expert panels.

6 Conclusion

This paper presented an AI-augmented Model-Based Systems Engineering (MBSE) framework aimed at improving semantic interoperability across heterogeneous Digital Twin systems. The proposed approach integrates Large Language Models to automatically generate SysML artifacts from natural language specifications, a rule based transformation to derive ontological representations, and ontology alignment mechanisms to enable cross system semantic reasoning.

Experimental results demonstrate that the framework is capable of generating syntactically valid SysML diagrams with high structural completeness across the four supported diagram types (Block Definition, State Machine, Activity, and Sequence diagrams). The SysML-to-ontology transformation consistently produces logically coherent ontologies, while the ontology alignment process significantly improves semantic correspondence quality compared to baseline matching approaches.

Despite these promising results, several limitations remain. The current implementation supports a restricted set of SysML diagram types, and the experimental evaluation was conducted on two industrial inspired systems within the port logistics domain. In addition, semantic correctness assessment partially relies on expert judgment, which may introduce subjectivity.

Future work will focus on extending the framework to additional SysML diagram types and application domains, refining evaluation protocols with larger expert panels, and investigating domain specific fine tuning of Large Language Models. Further integration with Industry 4.0 standards such as the Asset Administration Shell and RAMI 4.0 is also planned to strengthen compliance and practical adoption in industrial Digital Twin ecosystems.

7 Acknowledgements

The MAGPIE project is funded by the European Commission as part of the Horizon 2020 Green Deal calls under grant agreement number 101036594.

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